

Skin cancer diagnosis using convolutional neural networks for smartphone images: A comparative study

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ABSTRACT

In the past few years many researches investigated algorithms to diagnose skin cancer lesions where melanoma is the deadliest type of skin cancer. In this paper, we propose three different methods for binary classification (melanoma and nevus) using smartphone images since there is a lack in research on it. The applied methods depend on different versions of CNN architectures namely; Alex-net, Mobilenet-V2, and Resnet-50. These three methods are divided into two groups; one group is the three nets with transfer learning applied on the original data, and the second group is the same methods used for group one but applied on augmented data. The proposed methods are applied on PAD-UFES-20 which is a smartphone images dataset. The augmented data have improved the performance of each method relative to its performance for original data. The best performance is achieved when using Alex-net with transfer learning (TL) and data augmentation, as it achieved 99% for accuracy, 96.364% for sensitivity, 99.792% for specificity, 99.167% for precision, and 97.661% for F1-score. Mobilenet-V2 with TL on augmented dataset achieved promising results if it used as a mobile application as it achieved 94.071% for accuracy, 84.547% for sensitivity, 96.25% for specificity, 84.223% for precision, and 84.2% for F1-score. And finally, Resnet-50 with TL on augmented dataset achieved 94.918% for accuracy, 77.275% for sensitivity, 98.918% for specificity, 95.192% for precision, and 84.737% for F1-score.

1. Introduction

Skin cancer detection is an interesting and trending topic for researchers in the past few years, this is because skin cancer diagnosis has exacerbated dramatically in the past few years. Between 2009 and 2019 skin cancer was diagnosed with a percentage of 54% (Putra, Rufaida, & Leu, 2020). Melanoma is the most threatening and deadliest form of skin cancer, as it leads to more than 9000 deaths every year in the United States (Codella et al., 2017) (Yu et al., 2018). In Australia there are more than 13,000 new cases are annually diagnosed with melanoma which causes deaths to over 1200 cases. Every year in Europe, melanoma causes more than 20,000 deaths (Codella et al., 2017) (Begum & Asra, 2017).

Skin cancer can be cured if early detected, but only highly-trained specialists are capable of accurately diagnosing skin cancer early. There is a severe shortage of experts and tools that can medically diagnose skin cancer accurately, so the innovation of computer-aided systems that can detect skin cancer could help in saving more lives at

a low cost.

Most of the previous work in the detection of skin cancer concentrated on applying machine learning on dermoscopic images (images captured with dermoscopic devices in large clinics and hospitals). To the best of our knowledge, there is a shortage in applying new methods on smartphone images of skin cancer. Creating computer-aided diagnosis systems (CADs) for smartphone images of skin cancer can be used as an alarm for dangerous cases. Especially, when these CADs are used in places where workers deal with irradiation sources which is one of the main reasons for skin cancer infections. In such places, Mobile applications of such CADs on mobile phones may be a quick solution with a low cost.

In this study, it is suggested to create a CAD system that can classify images of skin cancer that are captured with smartphones into two classes melanoma class or nevus class. The proposed system consists of three models, three pre-trained networks which are Alex-net, Mobilenet-V2, and Resnet-50. Each network is used twice; one time with transfer learning while the other time with transfer learning and data

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augmentation.

2. Deep learning techniques

Alex-net was first proposed by (Krizhevsky, Sutskever, & Hinton, 2012), it has the advantages of reducing noise and enhancing features as every layer contains filters, this advantage reduces the need for pre-processing which reduces running time and reach high accuracy in less time. It has another essential advantage which is every convolutional layer in the architecture is followed by pooling layers that help in reducing features and only pass the essential features to the next layer.

Mobilenet-V2 has high performance when used in feature extraction for object segmentation or detection which is one of the main phases in skin cancer detection. As we deal with smartphone skin cancer images, in this paper, the proposed model presented will be used on mobile phones, so a framework that has small access for the main memory is needed. And this feature is presented in Mobilenet-V2 (Sandler, Howard, Zhu, Zhmoginov, & Chen, 2018).

Resnet-50 achieved high performance in image classification (Rusakovsky et al., 2015). As the Resnet-50 has the ability to learn complex features by adding more layers which help in solving complex problems, these problems lead at the end to improving accuracy. On the other hand, Resnet-50 achieved promising results in dermoscopic skin images (Shahin, Kamal, & Elattar, 2018).

Transfer learning enables the ability to use the knowledge gained from a previously learned problem and apply them to new problems. Here, in this paper, transfer learning is applied with three pre-trained networks which are Resnet50, Alex-net, and Mobilenet-V2.

Image data augmentation provides the ability to apply multiple pre-processing processes like resizing, rotation, and reflection. Data augmentation helps in overcoming the small datasets. According to previous researches, data augmentation increases the accuracy of the models (Ayan & Ünver, 2018) (Pham, Doucet, Luong, Tran, & Hoang, 2020).

The next sections in this paper are arranged as follows: the second section discusses the related work, the third section explains the proposed algorithms, the fourth section demonstrates the results and discussions and finally, the last section provides the conclusion and future work.

3. Related work

Many previous applications and methods are proposed in skin cancer detection and classification based on different techniques and methods.

3.1. Models used data pre-processing and augmentation

Colour and textures features are used in the model proposed in (Giotis et al., 2015) in order to classify the images into melanoma and benign. They applied Gaussian noise in the pre-processing, k-means algorithm is used in segmentation process then color and textures features are extracted and the last step they applied Cluster-based Adaptive Metric (CLAM) classifier in image classification. This model achieved an accuracy of 81% on the Med-node dataset.

Data augmentation is used in (Ayan & Ünver, 2018) in order to compare the results when the CNN model is trained twice, once with non-augmented data and the second time with augmented data. The model target is image classification of skin cancer into malignant and benign, the achieved accuracy for non-augmented data is 78% and for augmented data is 81%, the model is applied on the ISBI 2016 (Gutman et al., 2016).

Some researches applied other pre-processing techniques, in (Sri-vidhya, Sujatha, Ponnagall, Durgadevi, & Madheshwaran, 2020) edge detection and filtering are used then extract highly-perspective-features (shape, color, diameter, irregularities, and boundaries) then CNN was applied in malignant tumor detection with an accuracy of 95%. The

model is applied on (DermIS, October 2021) and (DermQuest, October 2021).

3.2. Models used deep learning directly

In (Codella et al., 2017), the researchers used a combination of deep learning features and hand-coded features. The proposed method targets detection and classification into two classes melanoma or nevi. The accuracy achieved is 76%. In this method, they used deep learning features with edge histogram, color histogram then U-net is applied in classification, the method is applied on The ISBI 2016 dataset (Gutman et al., 2016).

Some researches combine convolutional neural networks (CNN) with other classifiers like in (Albert et al., 2020). They proposed a method called Predict-Evaluate-Correct K-fold (PECK) to diagnose melanoma cancer with an accuracy of 91%. This PECK algorithm is based on merging support vector machine with deep convolutional neural networks. The model is applied on the MED-NODE dataset (Giotis et al., 2015).

Some other researches applied CNN only like the method used in (Winkler et al., 2020) which introduced an algorithm for image classification of skin cancer into melanoma and benign, they used convolutional neural networks and achieved an accuracy of 95%. The model is applied on different datasets that are collected from departments of dermatology in universities of Heidelberg, Munich, and Lyon.

Support vector machine (SVM) is used in the classification of skin cancer types in some researches (Waheed, Waheed, Zafar, & Riaz, 2017). Created a model to detect melanoma cancer with an accuracy of 96%. They extract color features and textures features then SVM is applied on the features vector in order to classify the images into melanoma or not melanoma. The model is applied on PH2 Dataset (Mendonça, Ferreira, Marques, Marcal, & Rozeira, 2013).

Other researches applied segmentation techniques before CNN such as in (Ashraf et al., 2020). In this research, k-means is applied to detect Region of Interest (ROI) then CNN with transfer learning and data augmentation is applied in the detection of melanoma cancer. This research achieved 97% accuracy. The model is applied on (DermIS, October 2021) and (DermQuest, 2021).

There is research that used the PAD-UFES-20 dataset, which is the dataset used in this paper. In the research (Moldovanu, Obreja, Biswas, & Moraru, 2021), the Feed forward Back Propagation Network (FFBPN) is used to classify skin cancer into malignant and benign, where the researchers applied the method on a hybrid dataset from four different dermoscopic and non-dermoscopic datasets. These hybrid four datasets are 7-Point (Kawahara, Daneshvar, Argenziano, & Hamarneh, 2018), Med-node (Giotis et al., 2015), PAD-UFES-20 (Pacheco et al., 2020) and PH2 (Mendonça et al., 2013). The accuracy achieved for all datasets is 96%. In this paper, it is difficult to compare the results achieved with the results in (Moldovanu et al., 2021), because they used a hybrid of four datasets and no mention for the accuracy of each dataset, on the other hand in this paper the method is applied on PAD-UFES-20 only.

In this paper, different pre-trained convolutional neural networks are applied on the melanoma detection problem. The nets used are Resnet50, Alex-net, and MobileNet-V2. Every net is applied two times, the first one is applied with transfer learning and the second time the net is applied with transfer learning and data augmentation. Data augmentation achieved high performance compared with methods without data augmentation as we worked with two classes (melanoma and nevus) with a limited number of images, so data augmentation creates different variations of images that can enlarge the size of the dataset.

4. Proposed model

In the proposed model, three pre-trained networks (Alex-net, MobileNet-V2, resnet50) are applied in two different ways, first with

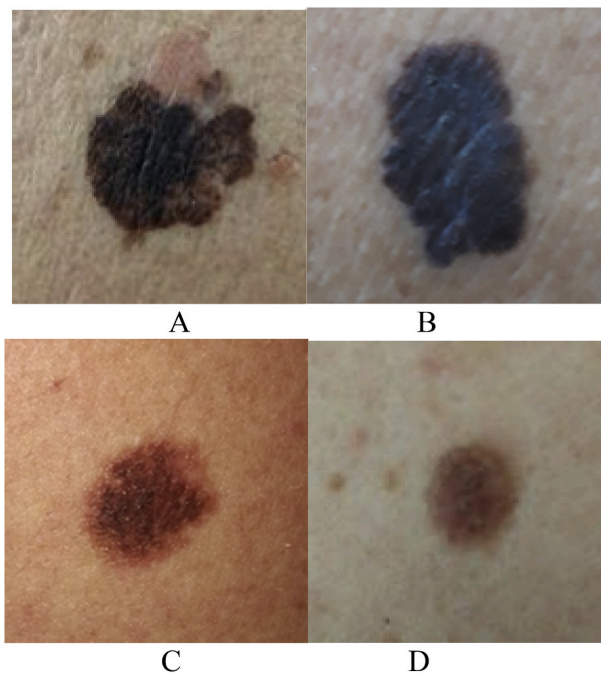


Fig. 1. Samples of images from PAD-UFES-20, A and B Melanoma, C and D Nevus.

transfer learning and second with transfer learning and data augmentation.

4.1. Pre-processing and dataset

In this research, the used dataset is PAD-UFES-20 which consists of 2298 images for six different skin cancer types (Pacheco et al., 2020). The images are collected from various smartphone devices, so the images have different sizes, resolutions, and lighting conditions.

In this study, two types of skin cancer of this dataset are used which are melanoma (MEL) and nevus (NEV). The number of images for each skin cancer type is 52 and 244 respectively, samples of the dataset images are shown in (Fig. 1).

Before the training process, all images are resized according to the input size of the used networks. In the case of Alex-net, the images are

resized to $227 \times 227 \times 3$, while in the case of Mobilenet-V2 and Resnet-50 the images are resized to $224 \times 224 \times 3$.

4.2. Pretrained networks

4.2.1. Alex-net with transfer learning

Alex-net consists of five convolutional layers, three fully connected layers, the softmax layer, and the output layer. In this research, the third fully connected layer is replaced with a new fully-connected layer with tuning the weights. The softmax layer is replaced with a new softmax layer, then the classification layer for 1000 classes of original Alex-net is replaced with a classification layer with two classes melanoma and nevus. Fig. 2 shows the diagram of the updated Alex-net with transfer learning.

4.2.2. MobileNet-V2 with transfer learning

MobileNet-V2 consists of the initial fully convolution layer, then 19 residual bottleneck layers, followed by convolution layer then average pooling layer, the next layer is a fully connected layer, followed by a softmax layer then at the end the output classification layer for 1000 classes (Sandler et al., 2018). In this research, transfer learning is applied and the last three layers are replaced with transformed layers for two classes, Fig. 3 shows the diagram of the updated MobileNet-V2 with transfer learning.

4.2.3. Resnet-50 with transfer learning

Resnet-50 architecture contains five phases, each phase contains two blocks where the first one is a convolution block and the second block is an identity block. Each convolution block consists of three convolution layers also each identity block has three convolution layers. In the end, there is a fully connected layer and a classification output layer which are replaced with three new layers for two classes when transfer learning is applied. Fig. 4 shows the diagram of the updated Resnet-50 with transfer learning.

4.2.4. Data augmentation

Data augmentation is added to our proposed model to prevent it from overfitting and to overcome the small number of training images. In this research, different augmentation methods are used.

The applied methods are random rotation with range $[-5, 5]$, random reflection with 50% probability, and random shearing in x and y with range $[-0.05, 0.05]$.

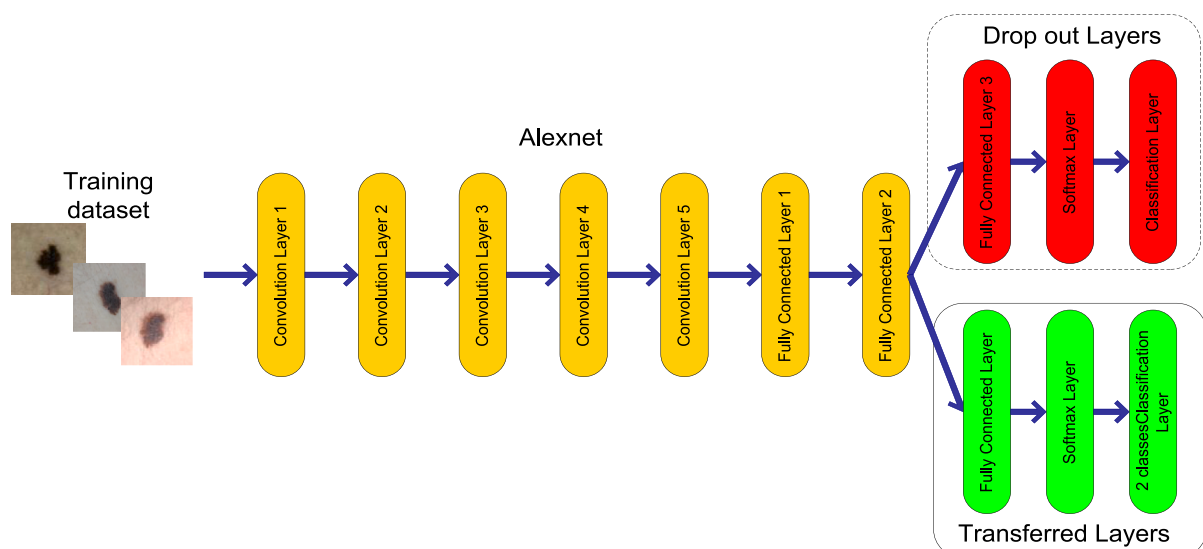


Fig. 2. Alex-net with transfer learning diagram.

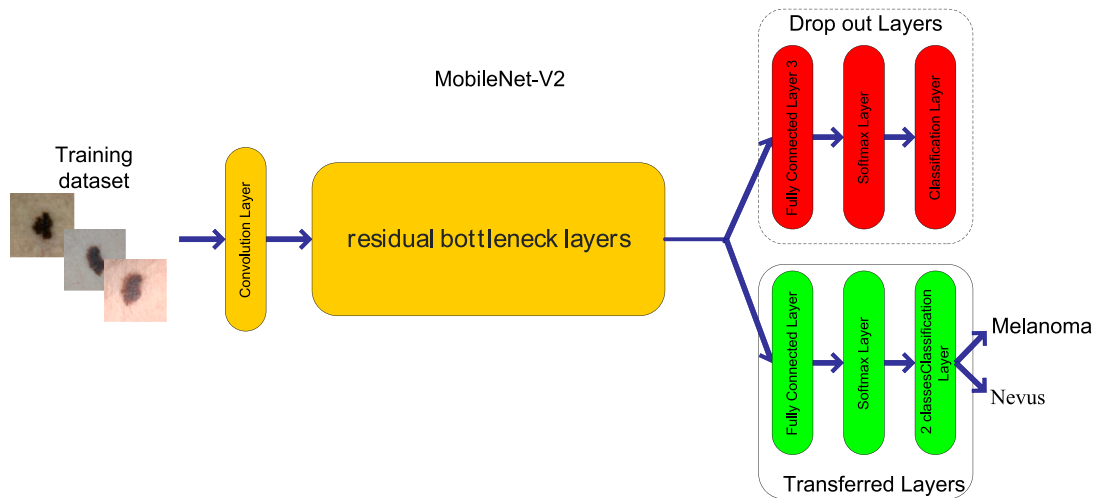


Fig. 3. MobileNet-V2 with transfer learning diagram.

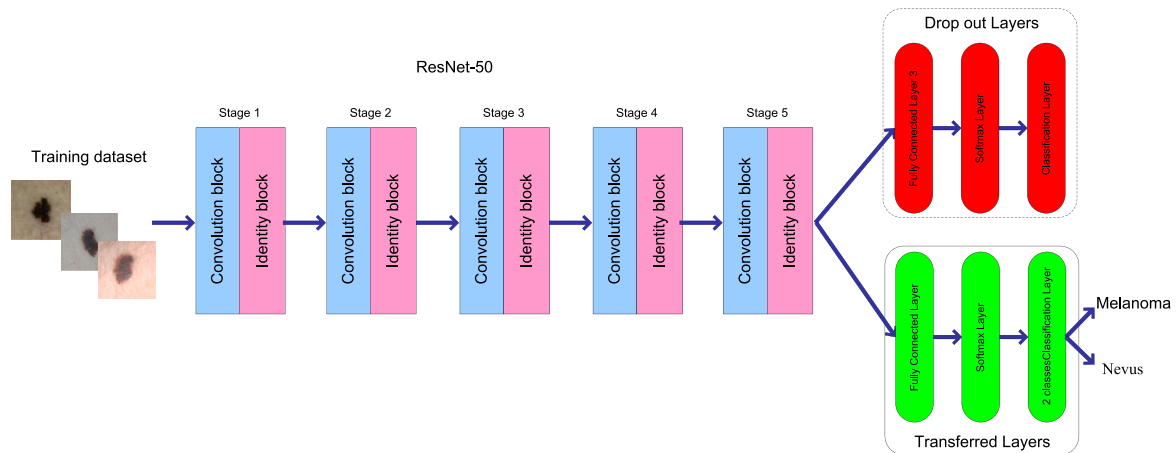


Fig. 4. Resnet-50 with transfer learning diagram.

5. Experimental results and discussions

This section will cover the training details and the results. In the training of the proposed model, the training options are constant for the three models. The training data is split into two parts, the first part is for training which is 80% of the overall data, while the second part is for testing and is 20% of the overall data. The optimizer used is 'adam', the number of epochs used is 50, the minibatch size used is 32, the initial learning rate used in this research is 0.0001, the used learn rate drop factor is 0.3, L2 regularization used value is 0.01.

The proposed methods are implemented on MATLAB 2019 64-bit. The system used has 16GB RAM and processor Intel with core i7 and Nvidia Geforce Gtx 1060 graphic card. All experiments used the same training options. The output results are compared based on accuracy when original data is used and when augmented data is used.

The performance measures used are accuracy, sensitivity (Recall),

specificity, precision (Stojanovi et al., 2014), and F1score (Adegun & Viriri, 2020). The following equations are used:

$$\text{Accuracy} \cdot = \frac{tp + tn}{tp + fp + fn + tn}$$

$$\text{Sensitivity} \cdot (\text{TPR}) \cdot = \frac{tp}{tp + fn}$$

$$\text{Specificity} \cdot (\text{TNR}) \cdot = \frac{tn}{fp + tn}$$

$$\text{Precision} \cdot (\text{PPV}) \cdot = \frac{tp}{tp + fp}$$

$$\text{F1Score} \cdot = \frac{2 \cdot (\text{Recall} \cdot \text{Precision})}{\text{Recall} + \text{Precision}}$$

Table-1

Evaluation measures of the proposed methods.

Pretrained networks with TL	Original dataset					Augmented dataset				
	Accuracy	sensitivity	specificity	precision	F1score	Accuracy	Sensitivity	specificity	precision	F1score
Alex-net	96.103	87.274	98.75	94.581	88.25	99.155	96.364	99.792	99.167	97.661
Mobilenet-V2	92.207	78.184	97.5	87.5	78.121	94.071	84.547	96.25	84.223	84.2
Resnet	93.392	75.457	99.375	96.111	78.9	94.918	77.275	98.959	95.192	84.737

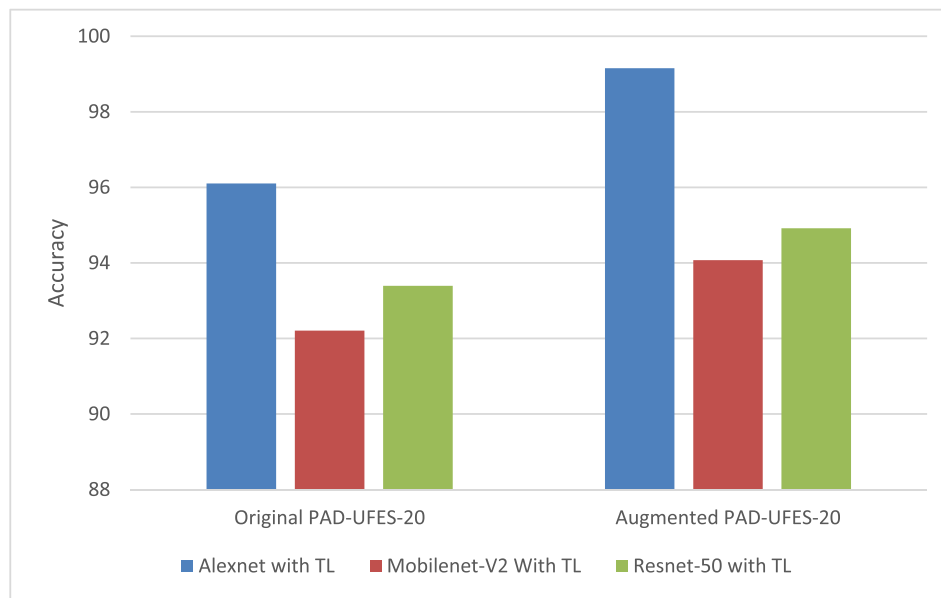


Fig. 5. Accuracy of compared methods for original and augmented PAD-UFES-20.

		Target class	
Output class		Mel	Nev
	Mel	9	0
	Nev	2	48
Alex-net with TL on the original dataset			

		Target class	
Output class		Mel	Nev
	Mel	10	0
	Nev	1	48
Alex-net with TL on the augmented dataset			

		Target class	
Output class		Mel	Nev
	Mel	9	2
	Nev	2	46
Mobilenet-V2 with TL on the original dataset			

		Target class	
Output class		Mel	Nev
	Mel	9	1
	Nev	2	47
Mobilenet-V2 with TL on the augmented dataset			

		Target class	
Output class		Mel	Nev
	Mel	7	0
	Nev	4	48
Resnet-50 with TL on the original dataset			

		Target class	
Output class		Mel	Nev
	Mel	8	0
	Nev	3	48
Resnet-50 with TL on the augmented dataset			

Fig. 6. Confusion matrix for the six different experiments.

where the variable tp refers to true positive, tn refers to true negative, fp refers to false positive, and fn refers to false negative. TPR indicates true positive rate, TNR indicates true negative rate and PPV indicates positive prediction value.

All proposed methods start by loading the training data, then

splitting the data into testing (20%) and training (80%). After that pass-through pre-processing step in order to resize images to the size of the pre-trained networks used in the methods. In the case of Alex-net, the size is converted to $227 \times 227 \times 3$, in Mobilenet-V2 and Resnet-50 the size is converted to $224 \times 224 \times 3$, then the pre-trained networks are applied. Data augmentation is applied only on the training data in order to improve the training process, but not applied on the testing data.

The process of the proposed methods is repeated 10 times, and the evaluation measures are computed with the average values for the 10 times.

Table 1 shows the results of the proposed methods when implemented on the original dataset and augmented dataset. The best results are achieved when Alex-net is used with transfer learning on the augmented dataset. It achieves 99% for Accuracy, 96% for Sensitivity, 99% for Specificity, 99% for Precision, and 97% for F1score. Confusion matrices for the proposed methods are shown in Fig. 6.

Comparison between the accuracies achieved between original and augmented PAD-UFES-20 are plotted in Fig. 5. Group one compares the results of the original PAD-UFES-20, the best accuracy achieved with Alex-net with TL 96.103%, group two compares the accuracies of the algorithms when applied on augmented PAD-UFES-20, and the best accuracy across them is Alex-net with TL which is 99.155%.

6. Conclusion and future work

This work provides methods to help dermatologists differentiate between melanoma and nevus using smartphone images. The research compares the results of three methods in two cases. Each method runs two times, once on the original dataset and the other time on the augmented dataset. The best results are achieved by Alex-net with transfer learning when trained by augmented images with an accuracy of 99.155%.

This research focuses on smartphone image classification, so if the method is used as a mobile application Mobilenet-V2 will be the appropriate method. When Mobilenet-V2 with transfer learning implemented on augmented data achieved promising results. The accuracy achieved by Mobilenet-V2 with transfer learning on the augmented dataset is 94.071%.

For future work, this research has some suggestions, first, we aim to classify seven types of skin cancer smartphone images. Second, we aim to improve the results of Mobilenet-V2 by applying different methods in

pre-processing including colour enhancement, removing noise, illumination effects, or extracting skin lesions and discarding non-lesion areas by applying suitable segmentation techniques. Third, it is suggested to apply pre-trained CNN networks that is not applied before in skin cancer images.

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